

61. If $\int_0^{\pi/3} \cos^4 x \, dx = a\pi + b\sqrt{3}$, where a and b are rational numbers, then $9a + 8b$ is equal to :

यदि $\int_0^{\pi/3} \cos^4 x \, dx = a\pi + b\sqrt{3}$, जहाँ a तथा b परिमेय संख्याएँ हैं, तो $9a + 8b$ बराबर होगा :

- (A) $\frac{3}{2}$ (B) 2 (C) 1 (D) 3

Ans. B

Sol. $\int_0^{\pi/3} \cos^4 x \, dx$

$$= \int_0^{\pi/3} \left(\frac{1 + \cos 2x}{2} \right)^2 dx$$

$$= \frac{1}{4} \int_0^{\pi/3} (1 + 2\cos 2x + \cos^2 2x) dx$$

$$= \frac{1}{4} \left[\int_0^{\pi/3} dx + 2 \int_0^{\pi/3} \cos 2x \, dx + \int_0^{\pi/3} \frac{1 + \cos 4x}{2} dx \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{3} + (\sin 2x)_0^{\pi/3} + \frac{1}{2} \left(\frac{\pi}{3} \right) + \frac{1}{8} (\sin 4x)_0^{\pi/3} \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{3} + (\sin 2x)_0^{\pi/3} + \frac{1}{2} \left(\frac{\pi}{3} \right) + \frac{1}{8} (\sin 4x)_0^{\pi/3} \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{2} + \frac{\sqrt{3}}{2} + \frac{1}{8} \times \left(-\frac{\sqrt{3}}{2} \right) \right]$$

$$= \frac{\pi}{2} + \frac{7\sqrt{3}}{64}$$

$$\therefore a = \frac{1}{8}; b = \frac{7}{64}$$

$$\therefore 9a + 8b = \frac{9}{8} + \frac{7}{8} = 2$$

62. Let f be a differentiable function such that $x^2 f(x) - x = 4 \int_0^x t f(t) dt$, $f(1) = \frac{2}{3}$. Then $18f(3)$ is equal to :

माना एक अवकलनीय फलन f के लिए $x^2 f(x) - x = 4 \int_0^x t f(t) dt$, $f(1) = \frac{2}{3}$ हैं। तो $18f(3)$ बराबर है :

(A) 180

(B) 210

(C) 160

(D) 150

Ans. C

Sol. Differentiate the given equation

$$\Rightarrow 2xf(x) + x^2 f'(x) - 1 = 4xf(x)$$

$$\Rightarrow x^2 \frac{dy}{dx} - 2xy = 1$$

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{2}{x}\right)y = \frac{1}{x^2}$$

$$\text{I.F.} = e^{\int -\frac{2}{x} dx} = \frac{1}{x^2}$$

$$\therefore y \left(\frac{1}{x^2} \right) = \int \frac{1}{x^4} dx$$

$$\Rightarrow \frac{y}{x^2} = \frac{-1}{3x^3} + c$$

$$\Rightarrow y = -\frac{1}{3x^3} + c$$

$$\Rightarrow y = -\frac{1}{3x} + cx^2$$

$$\therefore f(1) = \frac{2}{3} = -\frac{1}{3} + c \Rightarrow c = 1$$

$$f(x) = -\frac{1}{3x} + x^2$$

$$18f(3) = 160$$

63. Let f be a continuous function satisfying $\int_0^{t^2} (f(x) + x^2) dx = \frac{4}{3} t^3, \forall t > 0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to :

माना f एक सतत फलन है तथा $\int_0^{t^2} (f(x) + x^2) dx = \frac{4}{3} t^3, \forall t > 0$ है। तो $f\left(\frac{\pi^2}{4}\right)$ बराबर है :

(A) $\left(1 - \pi \left(1 + \frac{\pi^3}{16}\right)\right)$ (B) $-\pi \left(1 - \frac{\pi^2}{16}\right)$ (C) $\pi^2 \left(1 - \frac{\pi^3}{16}\right)$ (D) $\pi \left(1 - \frac{\pi^3}{16}\right)$

Ans. D

Sol. $\int_0^{t^2} (f(x) + x^2) dx = \frac{4}{3} t^3, \forall t > 0$

$$(f(t^2) + t^4) = 2t$$

$$f(t^2) = 2t - t^4$$

$$t = \frac{\pi}{2} \Rightarrow f\left(\frac{\pi^2}{4}\right) = \frac{2\pi}{2} - \frac{\pi^4}{16}$$

$$= \pi - \frac{\pi^4}{16} = \pi \left(1 - \frac{\pi^3}{16}\right)$$

64. The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x \left(\log_e \left(e^x + \sqrt{1 + e^{2x}} \right) \right) dx$ is equal to :

निम्न $\int_{-\log_e 2}^{\log_e 2} e^x \left(\log_e \left(e^x + \sqrt{1 + e^{2x}} \right) \right) dx$ का मान ज्ञात कीजिए :

(A) $\log_e \left(\frac{2(2 + \sqrt{5})}{\sqrt{1 + \sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$

(B) $\log_e \left(\frac{\sqrt{2}(2 + \sqrt{5})^2}{\sqrt{1 + \sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$

(C) $\log_e \left(\frac{(2 + \sqrt{5})^2}{\sqrt{1 + \sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$

(D) $\log_e \left(\frac{\sqrt{2}(3 - \sqrt{5})^2}{\sqrt{1 + \sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$

Ans. B

Sol. $I = \int_{-\log_e 2}^{\log_e 2} e^x \left[\log_e \left(e^x + \sqrt{1 + e^{2x}} \right) \right] dx$

Put $e^x = t$

$e^x dx = dt$

$$I = \int_{\frac{1}{2}}^2 1 \times \log_e \left[t + \sqrt{1 + t^2} \right] dt$$

$$= \left[t \ln \left(\sqrt{t^2 + 1} + t \right) \right]_{\frac{1}{2}}^2 - \int_{1/2}^2 \frac{t}{\sqrt{t^2 + 1}} dt$$

$$\begin{aligned}
 &= \left[t \ln \sqrt{t^2 + 1} + t - \sqrt{t^2 + 1} \right]_{\frac{1}{2}}^2 \\
 &= \left[2 \ln \sqrt{5} + 2 - \sqrt{5} \right] - \left[\frac{1}{2} \ln \left(\sqrt{\frac{5}{2}} + \frac{1}{2} \right) - \sqrt{\frac{5}{2}} \right] \\
 &= \ln \left(\frac{\sqrt{2}(2 + \sqrt{5})^2}{\sqrt{1 + \sqrt{5}}} \right) - \frac{\sqrt{5}}{2}
 \end{aligned}$$

65. Let the function $f : [0, 2] \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} e^{\min\{x^2, x - [x]\}}, & x \in [0, 1) \\ e^{[x - \log_e x]}, & x \in [1, 2] \end{cases}$ where $[t]$ denotes the greatest integer less than or equal to t . Then the value of the integral $\int_0^2 xf(x)dx$ is :

माना $f : [0, 2] \rightarrow \mathbb{R}$, $f(x) = \begin{cases} e^{\min\{x^2, x - [x]\}}, & x \in [0, 1) \\ e^{[x - \log_e x]}, & x \in [1, 2] \end{cases}$ द्वारा परिभाषित है, जहाँ $[t]$ का महत्तम पूर्णांक $\leq t$ है। तो समाकलन

$\int_0^2 xf(x)dx$ का मान है :

- (A) $2e - 1$ (B) $1 + \frac{3e}{2}$ (C) $2e - \frac{1}{2}$ (D) $(e - 1)\left(e^2 + \frac{1}{2}\right)$

Ans. C

Sol. $f(x) = \begin{cases} e^{x^2} & x \in (0, 1) \\ e & x \in [1, 2) \end{cases}$

Now $\int_0^2 xf(x)dx = \int_0^1 xe^{x^2} dx + \int_1^2 x \cdot e dx = 2e - \frac{1}{2}$

66. Let $[.]$ denote the greatest integer function. If $\int_0^{e^3} \left[\frac{1}{e^{x-1}} \right] dx = \alpha - \log_e 2$, then α^3 is equal to _____.

माना $[.]$ महत्तम पूर्णांक फलन है। यदि $\int_0^{e^3} \left[\frac{1}{e^{x-1}} \right] dx = \alpha - \log_e 2$ है तो α^3 बराबर है _____.

Ans. 8

Sol. $f(x) = \frac{1}{e^{x-1}} = e^{1-x}$

$$f(x) = 2$$

$$f(x) = 1$$

$$\frac{1}{e^{x-1}} = 2$$

$$x = 1$$

$$x = 1 - \ln 2$$

$$f(0) = e^1 = 2.71$$

$$f(e^3) = e^{1-e^3} \in (0, 1)$$

$$I = \int_0^{1-\ln 2} 2dx + \int_{1-\ln 2}^1 1dx + \int_1^{e^3} 0dx$$

$$= (1 - \ln 2 - 0) + 1(1 - 1 + \ln 2) + 0$$

$$\alpha = \ln 2 = 2 - \ln 2$$

$$\alpha = 2$$

$$\alpha^3 = 8$$

67. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a twice differentiable function. If for some $a \neq 0$, $\int_0^1 f(\lambda x) d\lambda = af(x)$, $f(1) = 1$

and $f(16) = \frac{1}{8}$, then $16 - f'\left(\frac{1}{16}\right)$ is equal to _____.

माना कि $f: (0, \infty) \rightarrow \mathbb{R}$, दो बार अवकलनीय फलन है। यदि किसी $a \neq 0$ के लिए $\int_0^1 f(\lambda x) d\lambda = af(x)$, $f(1) = 1$ और

$f(16) = \frac{1}{8}$ है, तो $16 - f'\left(\frac{1}{16}\right)$ बराबर है _____.

Ans. 112

Sol. $f: (0, \infty) \rightarrow \mathbb{R}$

$$\int_0^1 f(\lambda x) d\lambda = af(x)$$

$$f(1) = 1$$

$$\lambda x = t$$

$$x d\lambda = dt$$

$$\int_0^x f(t) \frac{dt}{x} = af(x)$$

$$\int_0^x f(t) dt = axf(x)$$

$$f'(x) = a(x f'(x) + f(x))$$

$$f(x) = axf'(x) + af(x)$$

$$f(x) = y$$

$$f'(x) = \frac{dy}{dx}$$

$$y = ax \frac{dy}{dx} + ay$$

$$(1-a)y = \frac{axdy}{dx}$$

$$\int (1-a) \frac{dx}{x} = \int a \frac{dy}{y}$$

$$(1-a)\ln x = a\ln y + C$$

$$(1-a)\ln x = a \ln f(x) + C$$

$$(1-a) \times 0 = a \times 0 + C$$

$$C = 0$$

$$(1-a)\ln x = a\ln f(x)$$

$$(1-a)\ln 6 = a \times \ln \left(\frac{1}{8} \right)$$

$$4(1-a)\ln 2 = -3a\ln 2$$

$$4\ln 2 - 4a\ln 2 = -3a\ln 2$$

$$4\ln 2 = a\ln 2$$

$$a = 4$$

$$-3\ln x = 4\ln f(x)$$

$$\ln f(x) = -\frac{3}{4} \ln x$$

$$f(x) = e^{-\frac{3}{4} \ln x}$$

$$f(x) = e^{\ln x - \frac{3}{4}}$$

$$f(x) = x^{-\frac{3}{4}}$$

$$f'(x) = -\frac{3}{4} \times x^{-\frac{3}{4}-1}$$

$$f'(x) = -\frac{3}{4} x^{-\frac{7}{4}}$$

$$f'\left(\frac{1}{16}\right) = -\frac{3}{4} \times \left(\frac{1}{16}\right)^{-\frac{7}{4}}$$

$$= -\frac{3}{4} \times 2^{-4 \times -\frac{7}{4}}$$

$$= -\frac{3}{4} \times 2^7$$

$$= -3 \times 2^5 = -96$$

$$16 - f'\left(\frac{1}{16}\right) = 16 + 96 = 112$$

68. If $\lim_{t \rightarrow 0} \left(\int_0^1 (3x+5)^t dx \right)^{\frac{1}{t}} = \frac{\alpha}{5e} \left(\frac{8}{5} \right)^{\frac{2}{3}}$, then α is equal to _____.

यदि $\lim_{t \rightarrow 0} \left(\int_0^1 (3x+5)^t dx \right)^{\frac{1}{t}} = \frac{\alpha}{5e} \left(\frac{8}{5} \right)^{\frac{2}{3}}$ है, तो α बराबर है _____.

Ans. 64

Sol. 1^∞ form

$$\text{Now } L = e^{\lim_{t \rightarrow 0} \frac{1}{t} \left(\frac{(3x+5)^{t+1}}{3(t+1)} \Big|_0^1 - 1 \right)}$$

$$= e^{\lim_{t \rightarrow 0} \frac{8^{t+1} - 5^{t+1} - 3t - 3}{3t(t+1)}}$$

$$= e^{\frac{8\ln 8 - 5\ln 5 - 3}{3}}$$

$$= \left(\frac{8}{5} \right)^{2/3} \left(\frac{64}{5} \right) = \frac{\alpha}{5e} \left(\frac{8}{5} \right)^{2/3}$$

On comparing

$$\alpha = 64$$

69. If $24 \int_0^{\pi/4} \left(\sin \left| 4x - \frac{\pi}{12} \right| + [2 \sin x] \right) dx = 2\pi + \alpha$, where $[.]$ denotes the greatest integer function, then α is equal to

यदि $24 \int_0^{\pi/4} \left(\sin \left| 4x - \frac{\pi}{12} \right| + [2 \sin x] \right) dx = 2\pi + \alpha$ जहाँ $[.]$ महत्तम पूर्णांक फलन निर्दिष्ट करता है, तो α बराबर है _____.

Ans. 12

Sol.

$$= 24 \int_0^{\pi/48} -\sin \left(4x - \frac{\pi}{12} \right) + \int_{\pi/48}^{\pi/4} \sin \left(4x - \frac{\pi}{12} \right) + \int_0^{\pi/6} [0] dx + \int_{\pi/6}^{\pi/4} \left[\underbrace{2 \sin x}_1 \right] dx$$

$$= 24 \left[\frac{\left(1 - \cos \frac{\pi}{12} \right)}{4} - \frac{\left(-\cos \frac{\pi}{12} - 1 \right)}{4} \right] + \frac{\pi}{4} - \frac{\pi}{6}$$

$$= 24 \left(\frac{1}{2} + \frac{\pi}{12} \right) = 2\pi + 12$$

$$\alpha = 12$$

70. If the shortest distance between the lines $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ and $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$, and $\int_0^k [x^2] dx = \alpha - \sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

यदि रेखाओं $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ तथा $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ के बीच न्यूनतम $\frac{38}{3\sqrt{5}}k$ है तथा $\int_0^k [x^2] dx = \alpha - \sqrt{\alpha}$ है, जहाँ $[x]$ महत्तम पूर्णांक फलन है, तो $6\alpha^3$ बराबर है _____

Ans. 48

Sol.

$$L_1 \equiv \frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$$

$$L_2 \equiv \frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$$

Shortest distance between L_1 and L_2 is $= \frac{\begin{vmatrix} 5 & 5 & -9 \\ 2 & 3 & 4 \\ 1 & -3 & 2 \end{vmatrix}}{\sqrt{(18)^2 + (0)^2 + (-9)^2}}$

$$\frac{38}{3\sqrt{5}}k = \frac{90+0+81}{9\sqrt{5}} = \frac{19}{\sqrt{5}}$$

$$\Rightarrow k = \frac{3}{2}$$

$$\text{Now } \int_0^k [x^2] dx = \int_0^{\frac{3}{2}} [x^2] dx$$

$$= \int_0^1 [x^2] dx + \int_1^{\sqrt{2}} [x^2] dx + \int_{\sqrt{2}}^{\frac{3}{2}} [x^2] dx$$

$$= 0 + (\sqrt{2} - 1) + 2\left(\frac{3}{2} - \sqrt{2}\right)$$

$$= 2 - \sqrt{2}$$

$$\Rightarrow \alpha = 2$$

$$\Rightarrow 6\alpha^3 = 48$$

71. If $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3} \right) + \frac{\pi}{b\sqrt{3}}$, where $a, b \in \mathbb{N}$, then $a + b$ is equal to _____.

यदि $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3} \right) + \frac{\pi}{b\sqrt{3}}$, जहाँ $a, b \in \mathbb{N}$ है, तो $a + b$ बराबर है _____

Ans. 8

Sol. $\ell = \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{\sin^2 x + \cos^2 x + \sin x \cos x} dx$

$$\ell = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x}{1 + \tan x + \tan^2 x} dx = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x \cdot \sec^2 x dx}{(1 + \tan^2 x)(1 + \tan x + \tan^2 x)}$$

Let $\tan x = t$

$$\ell = \int_0^1 \frac{t^2}{(1+t^2)(1+t+t^2)} dt$$

$$= \int_0^1 \left(\frac{x}{1+x^2} - \frac{x}{1+x+x^2} \right) dx$$

$$= \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx - \int_0^1 \frac{\frac{1}{2}(2x+1) - \frac{1}{2}}{1+x+x^2} dx$$

$$= \frac{1}{2} \ln 2 - \frac{1}{2} \ln 3 + \frac{1}{2} \int_0^1 \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{1}{2} \ln \frac{2}{3} + \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_0^1$$

$$= \frac{1}{2} \ln \frac{2}{3} + \frac{1}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right)$$

$$= \frac{1}{2} \ln \frac{2}{3} + \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$\therefore a = 2, b = 6$$

$$\therefore a + b = 8$$

72. If $f(t) = \int_0^\pi \frac{2x dx}{1 - \cos^2 t \sin^2 x}$, $0 < t < \pi$, then the value of $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ equals _____.

यदि $f(t) = \int_0^\pi \frac{2x dx}{1 - \cos^2 t \sin^2 x}$, $0 < t < \pi$ है, तो $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ का मान है _____

Ans. 1

Sol. $f(t) = \int_0^\pi \frac{2x dx}{1 - \cos^2 t \sin^2 x}$

$$x \rightarrow \pi - x$$

$$f(t) = \int_0^\pi \frac{2(\pi - x) dx}{1 - \cos^2 t \sin^2 x} = 2\pi \int_0^\pi \frac{dx}{1 - \cos^2 t \sin^2 x} - f(t)$$

$$\Rightarrow f(t) = \pi \int_0^\pi \frac{dx}{1 - \cos^2 t \sin^2 x}$$

$$= 2\pi \int_0^{\pi/2} \frac{dx}{1 - \cos^2 t \sin^2 x}$$

$$f(t) = 2\pi \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t \tan^2 x}$$

$$I_1 = \int \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t \tan^2 x}$$

$$\text{Put } \cos t \tan x = \lambda \Rightarrow \cos t \sec^2 x dx = d\lambda$$

$$I_1 = \int \frac{d\lambda}{\cos t (1 + \lambda^2 \sec^2 t - \lambda^2)} = \int \frac{d\lambda}{\cos t (1 + \lambda^2 \tan^2 t)}$$

$$= \frac{1}{\cos t \tan^2 t} \int \frac{d\lambda}{\lambda^2 + \cos^2 t} = \frac{1}{\cos t \tan^2 t} \times \frac{1}{\cos t} \tan^{-1}(\lambda \tan t)$$

$$= \frac{1}{\sin t} \tan^{-1}(\sin t \tan x)$$

$$\Rightarrow f(t) = \frac{2\pi}{\sin t} \tan^{-1}(\sin t \tan x) \Big|_0^{\frac{\pi}{2}} = \frac{\pi^2}{\sin t}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\pi^2 dt}{f(t)} = \int_0^{\frac{\pi}{2}} \sin t dt = 1$$

73. Let $r_k = \frac{\int_0^1 (1-x^7)^k dx}{\int_0^1 (1-x^7)^{k+1} dx}$, $k \in \mathbb{N}$. Then the value of $\sum_{k=1}^{10} \frac{1}{7(r_k - 1)}$ is equal to _____.

माना $r_k = \frac{\int_0^1 (1-x^7)^k dx}{\int_0^1 (1-x^7)^{k+1} dx}$, $k \in \mathbb{N}$ है। तो $\sum_{k=1}^{10} \frac{1}{7(r_k - 1)}$ का मान बराबर है _____

Ans. 65

Sol. $r_k = \frac{\int_0^1 (1-x^7)^k dx}{\int_0^1 (1-x^7)^{k+1} dx}$

$$\int_0^1 (1-x^7)^{k+1} dx = \int_0^1 (1-x^7)^k \cdot (1-x^7) dx$$

$$= \int_0^1 (1-x^7)^k dx - \int_0^1 x^7 (1-x^7)^k dx$$

$$\begin{aligned}
 &= \int_0^1 (1-x^7)^k - \int_0^1 x \cdot x^6 (1-x^7)^k dx \\
 &= \int_0^1 (1-x^7)^k - x \int_0^1 x^6 (1-x^7)^k dx + \int_0^1 \left[1 \cdot \int_0^1 x^6 \cdot (1-x^7)^k dx \right] dx \\
 &= \int_0^1 (1-x^7)^k - x \frac{(1-x^7)^{k+1}}{-7(k+1)} \Big|_0^1 + \int_0^1 \frac{(1-x^7)^{k+1}}{-7(k+1)} dx \\
 &\Rightarrow \int_0^1 (1-x^7)^{k+1} dx \left\{ 1 + \frac{1}{7(k+1)} \right\} = \int_0^1 (1-x^7)^k dx - 0
 \end{aligned}$$

$$\frac{7k+8}{7(k+1)} = \frac{\int_0^1 (1-x^7)^k dx}{\int_0^1 (1-x^7)^{k+1} dx}$$

$$\sum_{k=1}^{10} \frac{1}{7(r_k - 1)}$$

$$\Rightarrow \sum_{k=1}^{10} \frac{1}{7 \left(1 + \frac{1}{7k+7} - 1 \right)}$$

$$\Rightarrow \sum_{k=1}^{10} (K+1) \quad \Rightarrow \frac{10 \times 11}{2} + 10 \Rightarrow 55 + 10 = 65$$

74. Let $[t]$ denote the largest integer less than or equal to t . If

$$\int_0^3 \left([x^2] + \left[\frac{x^2}{2} \right] \right) dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}, \text{ where } a, b, c \in \mathbb{Z}, \text{ then } a + b + c \text{ is equal to } \underline{\hspace{2cm}}.$$

माना $[t]$, t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। यदि

$$\int_0^3 \left([x^2] + \left[\frac{x^2}{2} \right] \right) dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}, \text{ जहाँ } a, b, c \in \mathbb{Z} \text{ तो, } a + b + c \text{ बराबर है } \underline{\hspace{2cm}}$$

Ans. 23

Sol.
$$\int_0^3 \left([x^2] + \left[\frac{x^2}{2} \right] \right) dx$$

$$\begin{aligned}
 &= \int_0^1 0dx + \int_1^{12} 1dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx \\
 &+ \int_{\sqrt{3}}^2 3dx + \int_2^{\sqrt{5}} 4dx + \int_{\sqrt{5}}^{\sqrt{6}} 5dx + \int_{\sqrt{6}}^{\sqrt{7}} 6dx + \int_{\sqrt{7}}^{\sqrt{8}} 7dx + \int_{\sqrt{8}}^3 8dx + \int_0^{\sqrt{2}} 0dx + \int_{\sqrt{2}}^2 1dx \\
 &+ \int_2^{\sqrt{6}} 2dx + \int_{\sqrt{6}}^{\sqrt{8}} 3dx + \int_{\sqrt{8}}^3 4dx = 31 - 6\sqrt{2} - \sqrt{3} - \sqrt{5} - 2\sqrt{5}\sqrt{6} - \sqrt{7}
 \end{aligned}$$

$$a = 31, b = -6, c = -2$$

$$a + b + c = 31 - 6 - 2 = 23$$

75. Let $\lim_{n \rightarrow \infty} \left(\frac{n}{\sqrt{n^4+1}} - \frac{2n}{(n^2+1)\sqrt{n^4+1}} + \frac{n}{\sqrt{n^4+16}} - \frac{8}{(n^2+4)\sqrt{n^4+16}} + \dots + \frac{n}{\sqrt{n^4+n^4}} - \frac{2n \cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}} \right)$

be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to _____.

माना $\lim_{n \rightarrow \infty} \left(\frac{n}{\sqrt{n^4+1}} - \frac{2n}{(n^2+1)\sqrt{n^4+1}} + \frac{n}{\sqrt{n^4+16}} - \frac{8}{(n^2+4)\sqrt{n^4+16}} + \dots + \frac{n}{\sqrt{n^4+n^4}} - \frac{2n \cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}} \right) = \frac{\pi}{k}$

है, जब प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मान लिए गए हैं। तो k^2 बराबर है _____

Ans. 32

Sol. $\sum_{r=1}^{\infty} \frac{n}{\sqrt{n^4+r^4}} - \frac{2nr^2}{(n^2+r^2)\sqrt{n^4+r^4}}$

$$\sum_{r=1}^{\infty} \frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}} - \frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$$

$$\Rightarrow \int_0^1 \frac{dx}{\sqrt{1+x^4}} - \frac{2x^2 dx}{(1+x^2)\sqrt{1+x^4}}$$

$$\Rightarrow \int_0^1 \frac{1-x^2}{(1+x^2)\sqrt{1+x^4}} dx$$

$$\Rightarrow \int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}} dx$$

$$\Rightarrow -\int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}} dx$$

$$x + \frac{1}{x} = t \Rightarrow 1 - \frac{1}{x^2} dx = dt$$

$$\Rightarrow -\int_{\infty}^2 \frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow -\int_{\infty}^2 \frac{tdt}{t^2\sqrt{t^2-2}}$$

$$\text{take } t^2 - 2 = \alpha^2$$

$$tdt = \alpha d\alpha$$

$$\Rightarrow -\int_{\infty}^{\sqrt{2}} \frac{\alpha d\alpha}{(\alpha^2+2)\alpha}$$

$$\Rightarrow -\int_{\infty}^{\sqrt{2}} \frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow \frac{-1}{\sqrt{2}} \tan^{-1} \frac{\alpha}{\sqrt{2}} \Big|_{-\infty}^{\sqrt{2}}$$

$$\Rightarrow \frac{-1}{\sqrt{2}} \{\tan^{-1}\} + \frac{1}{\sqrt{2}} \tan^{-1} \infty$$

$$\Rightarrow \frac{1}{\sqrt{2}} \left\{ \frac{\pi}{2} - \frac{\pi}{4} \right\}$$

$$\Rightarrow \frac{\pi}{\sqrt{2}} = \frac{\pi}{K}$$

$$\text{So, } K = 4\sqrt{2}$$

$$K^2 = 32$$

76. Let $f(x) = \int_0^x g(t) \log_e \left(\frac{1-t}{1+t} \right) dt$, where g is a continuous odd function.

If $\int_{-\pi/2}^{\pi/2} \left(f(x) + \frac{x^2 \cos x}{1+e^x} \right) dx = \left(\frac{\pi}{\alpha} \right)^2 - \alpha$, then α is equal to _____.

माना $f(x) = \int_0^x g(t) \log_e \left(\frac{1-t}{1+t} \right) dt$ है, जहाँ g एक संतत विषम फलन है। यदि

यदि $\int_{-\pi/2}^{\pi/2} \left(f(x) + \frac{x^2 \cos x}{1+e^x} \right) dx = \left(\frac{\pi}{\alpha} \right)^2 - \alpha$ है, तो α बराबर है _____।

Ans. 2

Sol.
$$f(x) = \int_0^x g(t) \ln \left(\frac{1-t}{1+t} \right) dt$$

$$f(-x) = \int_0^{-x} g(t) \ln \left(\frac{1-t}{1+t} \right) dt$$

$$f(-x) = - \int_0^x g(-y) \ln \left(\frac{1+y}{1-y} \right) dy$$

$$= - \int_0^x g(y) \ln \left(\frac{1-y}{1+y} \right) dy \quad (g \text{ is odd})$$

$$f(-x) = -f(x) \Rightarrow f \text{ is also odd}$$

Now,

$$I = \int_{-\pi/2}^{\pi/2} \left(f(x) + \frac{x^2 \cos x}{1 + e^x} \right) dx$$

$$I = \int_{-\pi/2}^{\pi/2} \left(f(-x) + \frac{x^2 e^x \cos x}{1 + e^x} \right) dx$$

$$2I = \int_{-\pi/2}^{\pi/2} x^2 \cos x dx = 2 \int_0^{\pi/2} x^2 \cos x dx$$

$$I = \left(x^2 \sin x \right)_0^{\pi/2} - \int_0^{\pi/2} 2x \sin x dx$$

$$= \frac{\pi^2}{4} - 2 \left(-x \cos x + \int \cos x dx \right)_0^{\pi/2}$$

$$= \frac{\pi^2}{4} - 2(0 + 1) = \frac{\pi^2}{4} - 2 \Rightarrow \left(\frac{\pi}{2} \right)^2 - 2$$

$$\therefore a = 2$$

77. Let the slope of the line $45x + 5y + 3 = 0$ be $27r_1 + \frac{9r_2}{2}$ for some $r_1, r_2 \in \mathbb{R}$. Then

$$\lim_{x \rightarrow 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right) \text{ is equal to } \underline{\hspace{2cm}}.$$

माना किसी $r_1, r_2 \in \mathbb{R}$ के लिए रेखा $45x + 5y + 3 = 0$ की प्रवणता $27r_1 + \frac{9r_2}{2}$ है। तो $\lim_{x \rightarrow 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$

बराबर है $\underline{\hspace{2cm}}$.

Ans. 12

Sol. Slope of $45x + 5y + 3 = 0$ is $m = -9$

$$\therefore 27r_1 + \frac{9r_2}{2} = -9$$

$$\lim_{x \rightarrow 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$$

$$\begin{aligned} & \lim_{x \rightarrow 3} \frac{\int_3^x 8t^2 dt}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} \\ &= \lim_{x \rightarrow 3} \frac{8x^2}{\frac{3r_2}{2} - 2r_2 x - 3r_1 x^2 - 3} \text{ by L.H. rule} \\ &= \frac{8 \times 9}{\frac{3r_2}{2} - 6r_2 - 27r_1 - 3} \\ &= \frac{8 \times 9}{-\frac{9r_2}{2} - 27r_1 - 3} \\ &= \frac{8 \times 9}{6} = 12 \end{aligned}$$

78. If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x} dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$, where α , β and γ are rational numbers, then $3\alpha + 4\beta - \gamma$ is equal to _____.

यदि $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x} dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$, जहाँ α , β तथा γ परिमेय संख्याएँ हैं, तो $3\alpha + 4\beta - \gamma$ बराबर है _____.

Ans. 6

Sol.

$$\begin{aligned} &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x} dx \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} |\sin x - \cos x| dx \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\cos x - \sin x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sin x - \cos x) dx \\ &= -1 + 2\sqrt{2} - \sqrt{3} \end{aligned}$$

$$= \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$$

$$\alpha = -1, \beta = 2, \gamma = -1$$

$$3\alpha + 4\beta - \gamma = 6$$

79. The value of $9 \int_0^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx$, where $[t]$ denotes the greatest integer less than or equal to t , is _____.

$$9 \int_0^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx, \text{ जहाँ } [t] \text{ महत्तम पूर्णांक } \leq t \text{ है, का मान है } \underline{\hspace{2cm}} \text{।}$$

Ans. 155

Sol. $9 \int_0^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx$

$$\text{Let } I = \int_0^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx$$

$$\frac{10x}{x+1} = 1 \Rightarrow x = \frac{1}{9}$$

$$\frac{10x}{x+1} = 4 \Rightarrow x = \frac{2}{3}$$

$$\frac{10x}{x+1} = 9 \Rightarrow x = 9$$

$$I = \int_0^{\frac{1}{9}} 0 dx + \int_{\frac{1}{9}}^{\frac{2}{3}} 1 dx + \int_{\frac{2}{3}}^9 2 dx$$

$$= 0 + \left[x \right]_{\frac{1}{9}}^{\frac{2}{3}} + \left[2x \right]_{\frac{2}{3}}^9$$

$$I = \frac{2}{3} - \frac{1}{9} + 18 - \frac{4}{3}$$

$$\Rightarrow I = \frac{155}{9}$$

$$\Rightarrow 9I = 155$$

80. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{4^x}{4^x + 2}$ and $M = \int_{f(a)}^{f(1-a)} x \sin^4(x(1-x)) dx$,

$N = \int_{f(a)}^{f(1-a)} \sin^4(x(1-x)) dx$; $a \neq \frac{1}{2}$. If $\alpha M = \beta N$, $\alpha, \beta \in \mathbb{N}$, then the least value of $\alpha^2 + \beta^2$ is equal to _____.

माना फलन $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{4^x}{4^x + 2}$ द्वारा परिभाषित है तथा $M = \int_{f(a)}^{f(1-a)} x \sin^4(x(1-x)) dx$,

$N = \int_{f(a)}^{f(1-a)} \sin^4(x(1-x)) dx$; $a \neq \frac{1}{2}$ हैं। यदि $\alpha M = \beta N$, $\alpha, \beta \in \mathbb{N}$ हैं, तो $\alpha^2 + \beta^2$ का न्यूनतम मान है _____.

Ans. 5

Sol. $f(x) + f(1-x) = \frac{4^x}{4^x + 2} + \frac{4^{1-x}}{4^{1-x} + 2}$

$$= \frac{4^x}{4^x + 2} + \frac{4}{4 + 2 \cdot 4^x}$$

$$= \frac{4^x + 2}{4^x + 2} = 1$$

$$\Rightarrow M = \int_{f(a)}^{f(1-a)} x \sin^4(x(1-x)) dx$$

$$x \rightarrow f(a) + f(1-a) - x = 1 - x$$

$$M = \int_{f(a)}^{f(1-a)} (1-x) \sin^4((1-x)x) dx$$

$$M = N - M \Rightarrow \frac{M}{N} = \frac{1}{2} = \frac{\beta}{\alpha}$$

$$\Rightarrow (\alpha^2 + \beta^2)_{\text{least}} = 1 + 4 = 5$$